INFORMATION COMMUNICATION TECHNOLOGY

87%

Object-Oriented Programming in C++
Module 4
CAT 3

Time: 1 hour 30 minutes | **Total:** 40 marks

Candidate Name: BRIAN

Admission Number: _____ **Date:** _____

Instructions

1. Answer ALL questions.
2. Write your name and admission number in the spaces provided above.
3. Write all code neatly and use proper C++ syntax.

QUESTIONS

1. Define **object-oriented programming** and state what an object bundles together. (2 marks)
2. Explain the difference between a **class** and an **object**, using a real-world example. (3 marks)
3. Compare **procedural programming** and **object-oriented programming** in terms of focus, handling of data, and reuse. (3 marks)
4. Explain the OOP concepts of **encapsulation** and **polymorphism**, giving one example of each. (4 marks)

5. State **two advantages** and **two disadvantages** of object-oriented programming. (4 marks)
6. State the purpose of each of the following parts of a C++ program: (4 marks)

   a) #include <iostream>

   b) using namespace std;

   c) int main()

   d) return 0;

7. The program below contains errors. Identify **four** errors and write the corrected statements. (4 marks)
8. #include <iostream>
9. using namespace std
10.
11. int main() {
12. int age;
13. cin << age;
14. cout >> "Age is " >> age;
15. return 0;
16. }
17. a) Describe the three C++ access specifiers: **public**, **private** and **protected**. (3 marks)

    b) A programmer declares the balance attribute of a BankAccount class as **public**. Evaluate this decision and recommend a better approach. (3 marks)

18. a) Write a C++ class named Book with a string attribute title, an int attribute price, and a method display() that prints both values. All members should be public. (3 marks)

    b) Write the statements in main() that create an object of Book, assign values to its attributes, and call the display() method. (2 marks)

19. Write a complete C++ program that defines a class Rectangle with a method calculateArea(int length, int width) that returns the area. In main(), create an object, call

the method with values of your choice, store the result in a variable, and display it. (5 marks)

1. Is a Programming (OOP) is a Programming approach that Organizes a Program around Classes and Objects.

   An Object bundles together
   Data (attributes): The Properties of an Object.
   Methods (functions):

2.

Class	Object
Blue print template used to Create Objects.	An instance of a Class

Example

```
Class Car {
 public:
  String Colour;
 };
Car Car1; Object 1
Car Car2; object 2.
```

3.

	Procedural Programming	Object-Oriented Programming
focus	functions & Procedures	Classes and objects.
coding data	Data and functions are generally handled seperately	Data and Methods can be bundled together in Objects
Reuse	Reuses functions & Procedures	Reuses Code through Inheritance and other features.

Question 4]:

a] Encapsulation → is the process of Combining data and methods in one Class while Controlling access to the data.

e.g A Bank A/c keeps it balance Private and allow access through methods.

b] Polymorphism → means one interface Can have the Same name but accept different parameters

e.g. Two functions Can have the same name but accept different Parameters

c] Advantages

Code Reusability

Data Security.

Disadvantages

Complexity

Memory usage.

d] #include <iostream> → allows the program to use input and Output features such as Cin and Cout.

e] using namespace std;

allows the Programmer to use Standard C++ names such as Cout and Cin. without writing std:: before them.

f] int main()

Defines the main function where Program execution begins

g] return 0;

Indicates that the program has Completed Successfully.

Correct the following program

```
#include <iostream>
using namespace std;

int main()
{
    int num;
    cout << "Enter a number: ";
    cin >> num;
    if (num % 2 == 0)
        cout << num << " is even." << endl;
    else
        cout << num << " is odd." << endl;
    return 0;
}
```

Forced error

(a) Public
members can be accessed from outside the class through an object, subject to the level access rules

Private
members can be accessed within the class and by friends

Protected
Members can be accessed within the class, its friends and derived classes, subject to the usual access rules

(b) Declaring the Balance as Public is not recommended because other parts of the program can change it directly causing inconsistencies.

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18 #include <iostream>
#include <string>
using namespace std;

Class Book {
public:
string title;
int price;
Void display() {
Cout << "Title:" << title << endL;
Cout << "Price:" << price << endL;
};
};

??

(5) int main() {
Book b1;
b1.title = "introduction to CTT";
b1.price = 500;
b1.display();
return 0;
}

19 #include <iostream>
using namespace std;
Class Rectangle {
public:
int Calculate area (int Length, int Width) {
return length * width;
};
int main() {
Rectangle r1;
int result = r1.Calculatearea(10, 5);
Cout << "Area of rectangle:" << result << endL;
return 0;
}

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